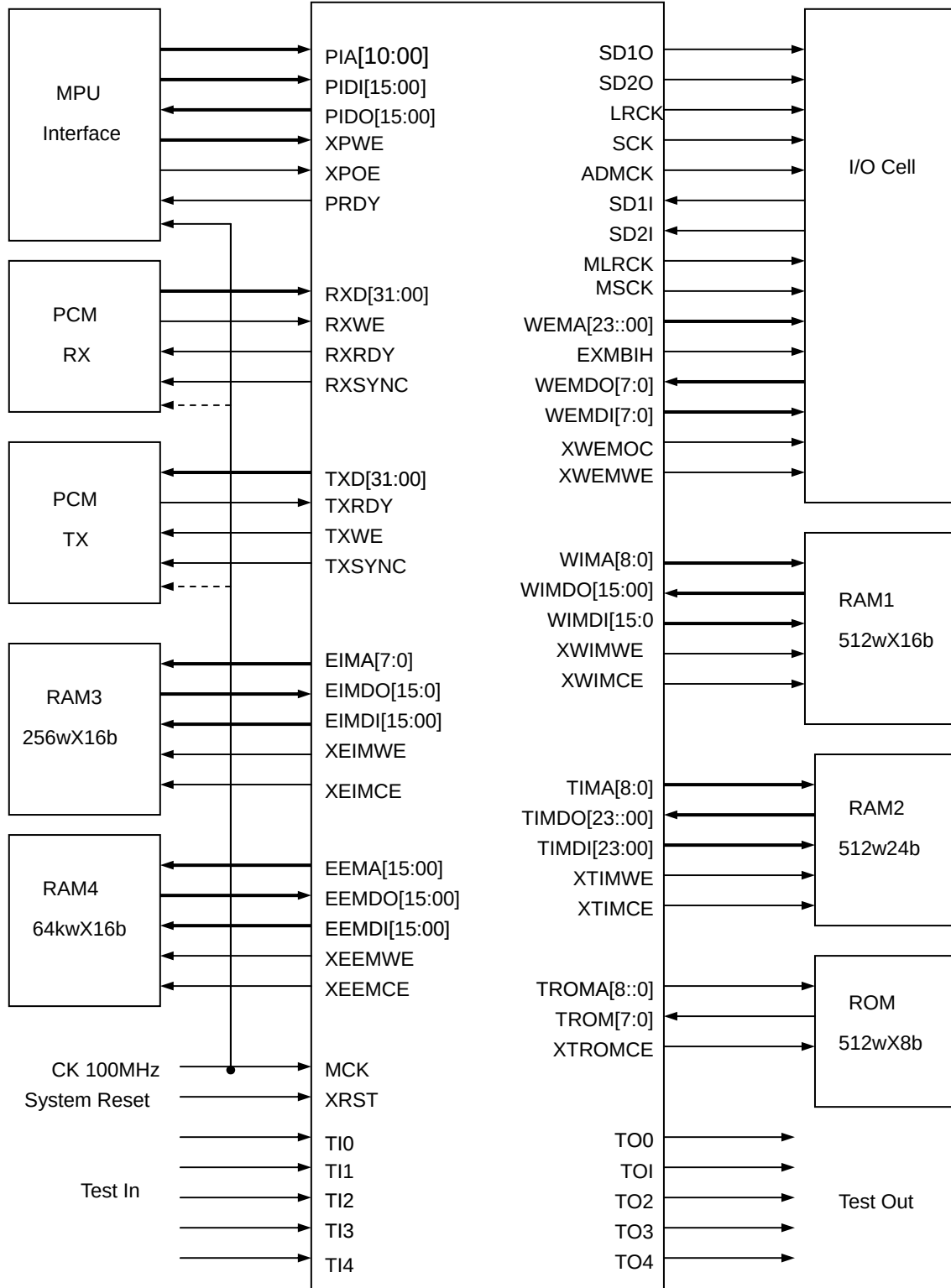


SoundEngine Block Conection

2007/2/19

SSI co/Yano

Sound Engine Block "SEIP1.5"



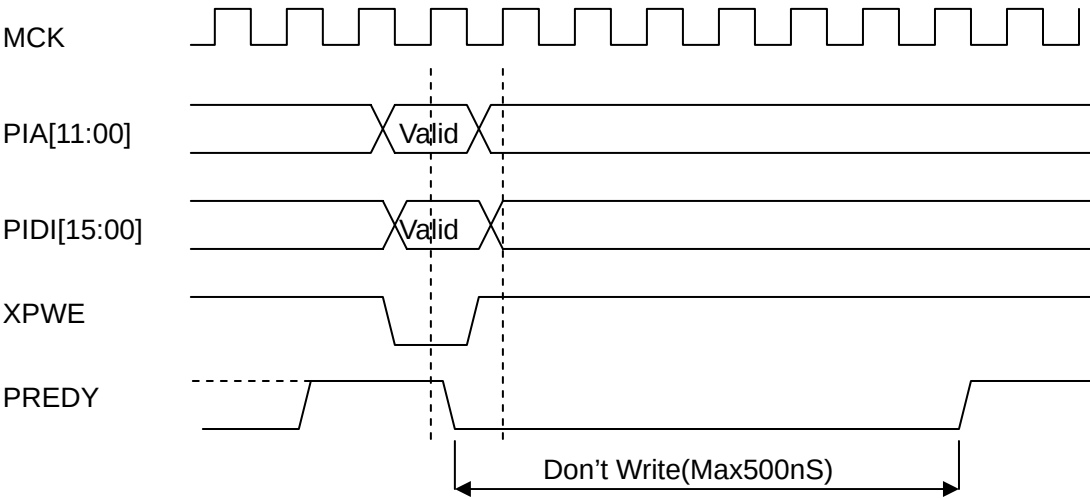
Pin Name	I/O	Polarity	
[System]-			
MCK	I	P	100MHz
XRST	I	L	System Reset
[MPU Interface]			
PIA[10:00]	I	D	SE Resister Address
PID[15:00]	I	D	SE Resister Write Data
PIDO[15:00]	O	D	SE Resister Read Data (Not All)
XPWS	I	L	SE Resister Write Select
XPOE	I	L	SE Resister Reed Select
RRDY	O	H	SE Resister Access Enable
[PCM RX]			
RXD[31:00]	I	D	PCM Rx Data[31:00] Lch[31:16] Rch[15:00]
RXWE	I	H	PCM Rx Data Write Enable
RXRDY	O	H	PCM Rx Ready
RXSYNC	O	H	PCM Rx Sync Clock
[PCM TX]			
TXD[31:00]	O	D	PCM Tx Data[31:00] Lch[31:15] Rch[15:00]
TXWE	O	H	PCM Tx Data Write Enable
TXRDY	I	H	PCM Tx Ready
TXSYNC	O	H	PCM Tx Sync Clock
[Inner RAM1 Interface]			Synchronous SRAM 512wX16b
WIMA[8:0]	O	D	Inner Memory1 Address[8:0]
WIMDO[15:00]	I	D	Inner Memory1 Reed Data[15:00]
XEIMWE	O	L	Inner Memory1 Write Enable
XEIMCE	O	L	Inner Memory1 Chip Enable
[Inner RAM2 Interface]			Synchronous SRAM 512wX24b
TIMA[8:0]	O	D	Inner Memory2 Address[8:0]
TIMDO[23:00]	I	D	Inner Memory2 Read Data[23:00]
TIMDI[23:00]	O	D	Inner Memory2 Write Data[23:00]
XTIMWE	O	L	Inner Memory Write Enable
XTIMCE	O	L	Inner Memory Chip Enable
{Inner RAM3 Interface}			Synchronous SRAM 256wX16b
EIMA[7:0]	O	D	Inner Memory3 Address[7:0]
EIMDO[15:00]	I	D	Inner Memory3 Read Data[15:00]
EIMDI[15:00]	O	D	Inner Memory3 Write Data[15:00]
XEIMWE	O	L	Inner Memory Write Enable
XEIMCE	O	L	Inner Memory Chip Enable
[Inner RAM4 Interface]			Synchronous SRAM 64KwX16b
EEMA[15:00]	O	D	Inner Memory4 Address[15:00]
EEMDO[15:00]	I	D	Inner Memory4 Read Data[15:00]
EEMDI[15:00]	O	D	Inner Memory4 Write Data[15:00]
XEEMWE	O	L	Inner Memory4 Write Enable
XEEMCE	O	L	Inner Memory4 Chip Enable
[Inner ROM Interface]			Synchronous ROM 512wX8b
TROMA[8:0]	O	D	Inner Memory5 Address[8:0]
TROMDO[7:0]	I	D	Inner Memory5 Read Data[7:0]
TROMCE	O	L	Inner Memory5 Chip Enable
[Extra Memory Interface]Chip I/O			Flash or Mask Memory 8MwX8b or 16MwX8b
WEMA[23:00]	O	D	Extra Memory Address[23:00]
EXMBIH	O	L	Extra Memory Address Control
WEMDO[7:0]	I	D	Extra Memory Read Data[7:0]
WEMDI[7:0]	O	D	Extra Memory Write Data[7:0]
XWEMWE	O	L	Extra Memory Write Enable
XWEMOC	O	L	Extra Memory Out Control(IO Control)
[D/A Converter Interfase]Chio I/O			
SD1O	O	D	Serial Data 1 Out
SD2O	O	D	Serial Data 2 Out
LRCKO	O	D	LRCK Out
SCKO	O	P	SCLK

[A/D Converter Interface]Chio I/O			
SD1I	I	D	Serial Data 1 Input
SD2O	I	D	Serial Data 2 Input
MLRCK	O	D	Mic LRCK Out
MSCK	O	P	Mic SCK Out
ADMCK	O	P	D/A A/D Master Clock
[Scan Test]			
TI0-TI4	I	D	
TE	I	L	
TO0-TO4	O		

Polarity H:High active L:Low active P:Positive edge N: Negative edge D:Data

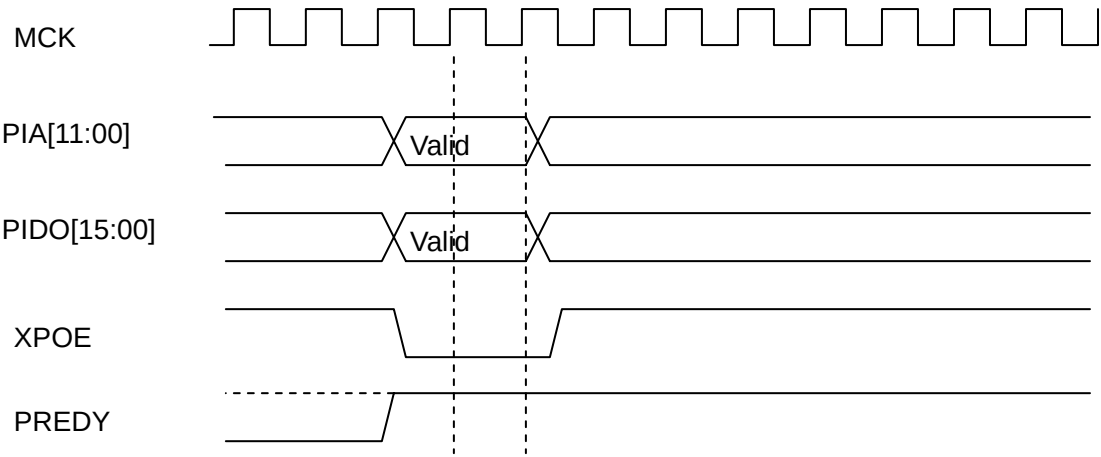
MPU InterFace

<MPU Write>



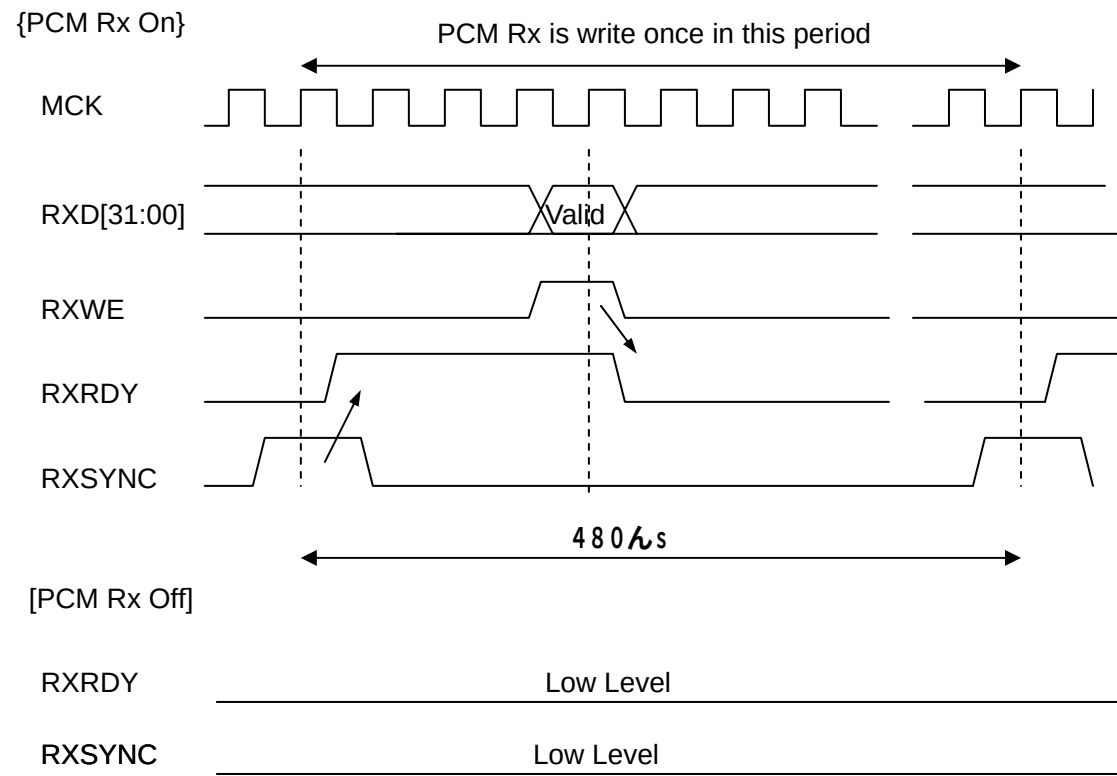
*XPOE=H

<MPU Read>

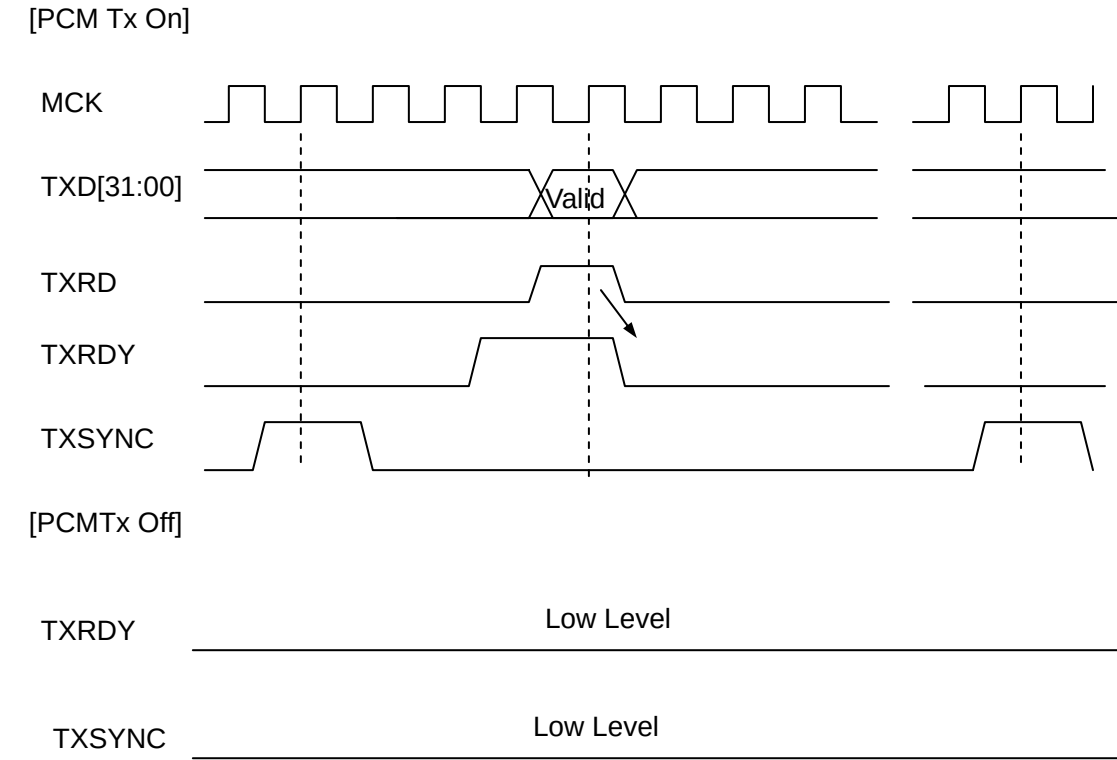


*XPWE=H

PCM RX Interface

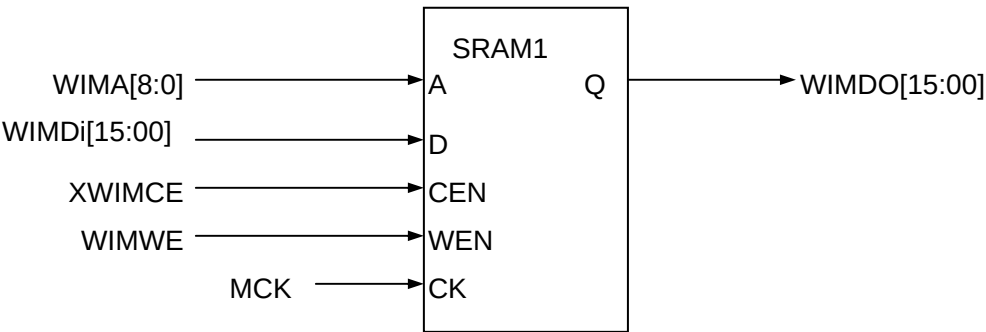


PCM TX Interface

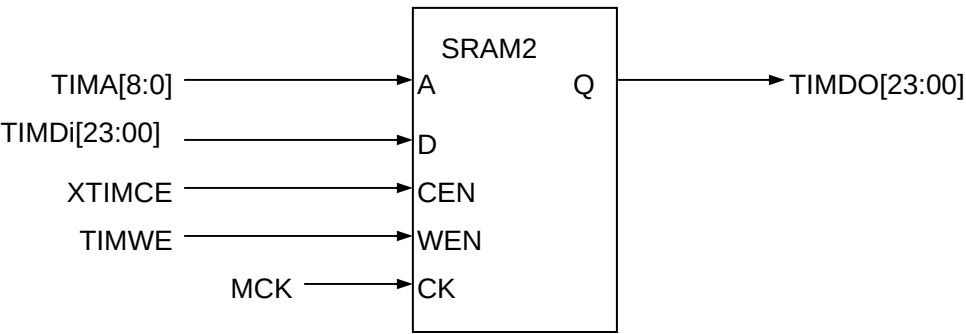


Inner SRAM Interface

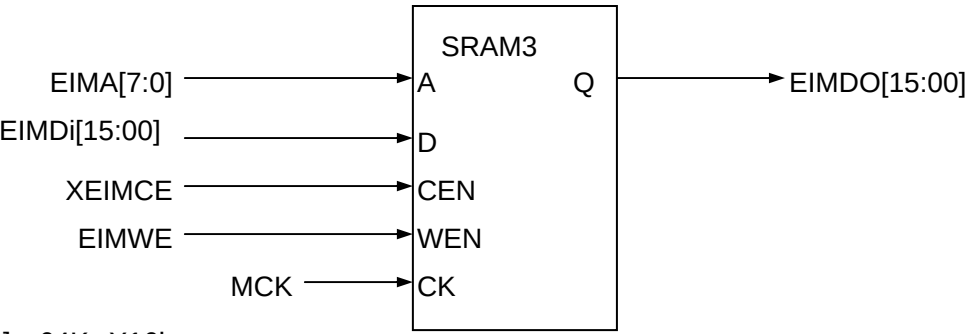
[SRAM1] 512wX16b



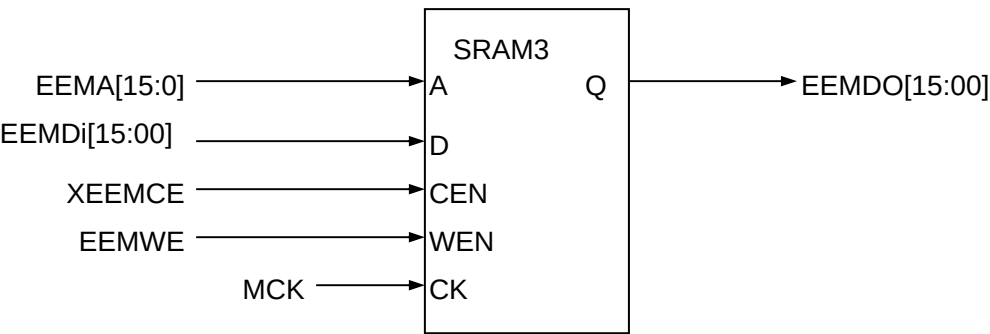
[SRAM2] 521wX24b



[SRAM3] 256wX16b

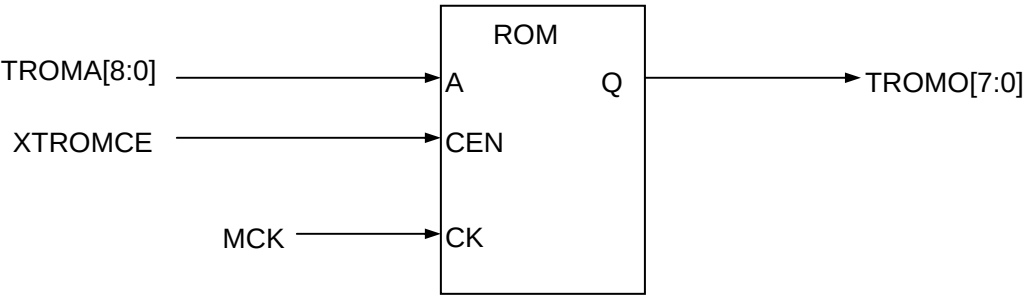


[SRAM4] 64KwX16b



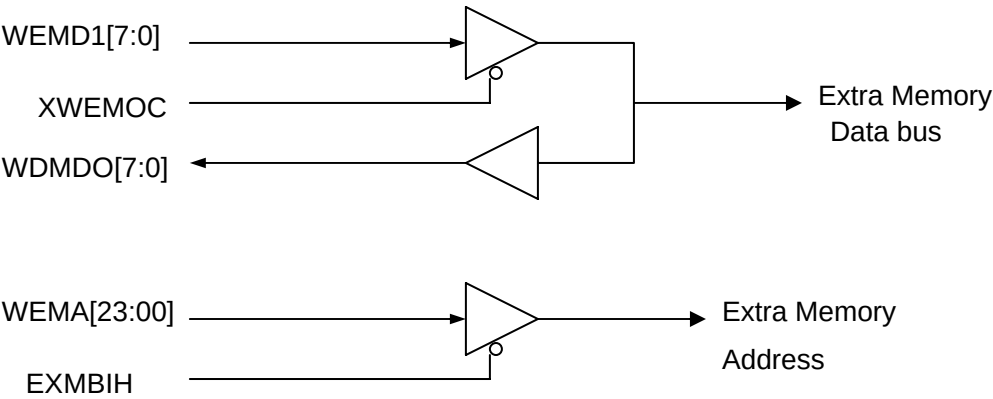
Inner ROM Interface

ROM 512wX8b



I/O Interface

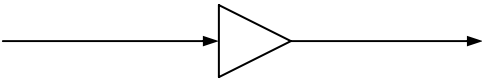
[Ext Memory]



[Others]

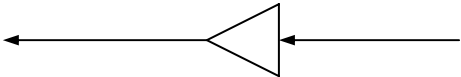
Output

Output Buf



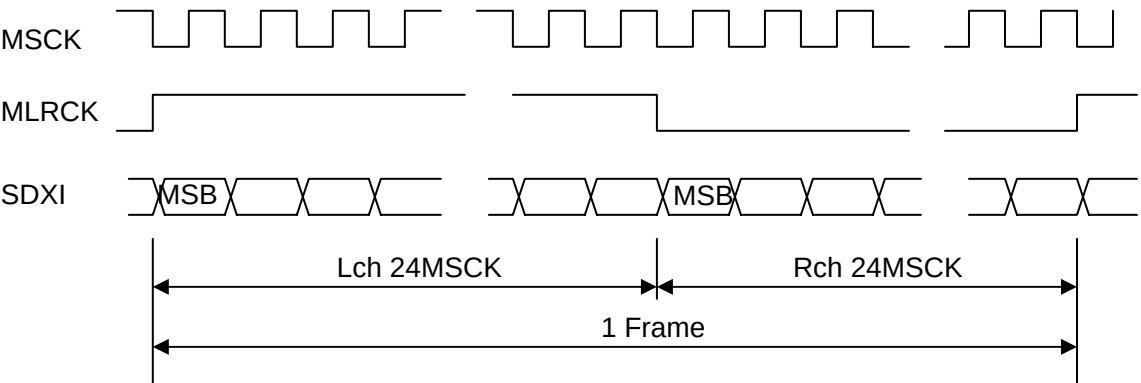
Input

Input Buf

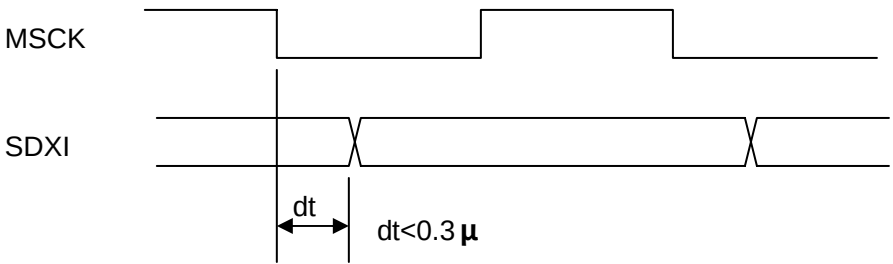


Simulation Input Timing

SD1I,SD2I(24bit 2ch SERIAL Input)



Sirial Data bit Delay



Flash MEMORY Access tTime

